

Project 1 ECE 456

Achilleas Lydakis

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1 Introduction

This project focuses on the complete implementation of the Manchester carry chain adder, starting from schematic design, moving through simulation, and followed by detailed analysis including power consumption, propagation delay measurement, and throughput calculation. The study concludes with a comparative evaluation, highlighting the performance of the Manchester adder against other common adder architectures. Through this process, key insights into the efficiency and design trade-offs of the Manchester carry chain are explored.

2 Building Blocks

The implementation begins with the creation of the fundamental building blocks required for the Manchester carry chain adder. Specifically, a XOR gate, an AND gate, and an OR gate are designed first. These basic logic gates serve as the core components that are later used to construct the full adder circuitry and ultimately the complete Manchester adder structure.

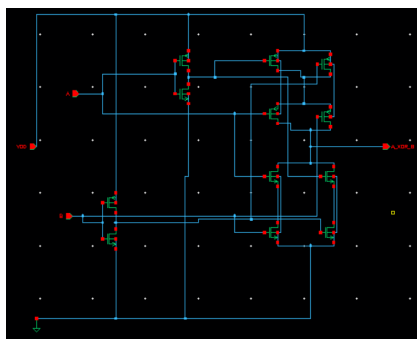


Figure 1: XOR Gate Schematic

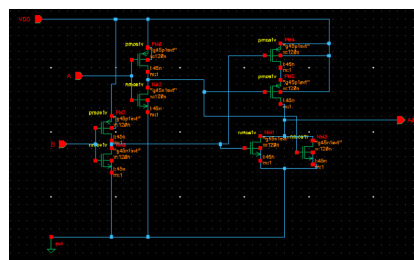


Figure 2: AND Gate Schematic

The AND gate is implemented by placing two inverters at the input signals, followed by a NOR gate, effectively creating an AND behavior using CMOS

logic principles. For the XOR gate, the standard CMOS implementation is used: first, the inputs are inverted using two inverters, and then the normal and inverted inputs are combined through appropriate pull-up and pull-down networks to generate the XOR function.

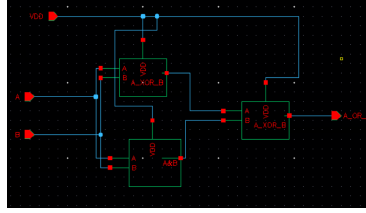


Figure 3: OR Gate Schematic

The OR gate is implemented by using the XOR and AND gates, combining their outputs appropriately to realize the OR functionality.

3 Final Design

Using the previously constructed XOR, AND, and OR gates, we build the complete Manchester carry chain adder. The final design integrates all required components into a single schematic that realizes the full adder functionality.

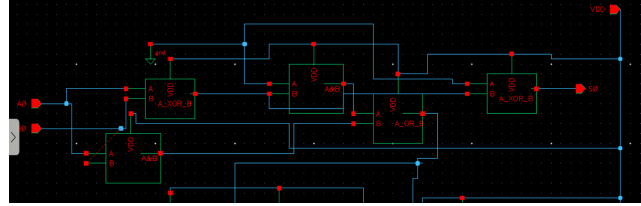


Figure 4: Final Design - Zoomed-In View of Logic Components

In the final design, for every input bit pair a_i and b_i , two intermediate signals are generated:

- p_i (propagate) using a XOR gate: $p_i = a_i \oplus b_i$
- g_i (generate) using an AND gate: $g_i = a_i \cdot b_i$

These signals are then used to create the carry-out signal c_{i+1} and the sum output s_i for each bit. The carry c_{i+1} is formed by combining the propagate and generate signals through appropriate logic gates, while the sum s_i is generated by XOR-ing p_i with the carry-in c_i .

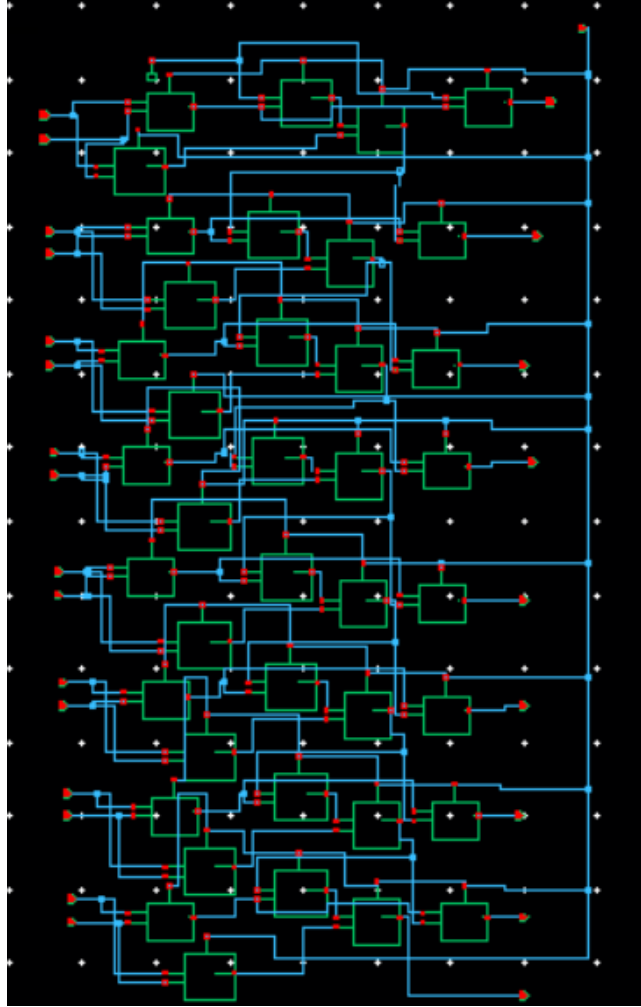


Figure 5: Final Design - Full Schematic Overview

To simplify simulation and power analysis, the final schematic was reorganized into a more compact layout. This optimized version reduces wire complexity and area usage, making the simulation setup and power measurement process more efficient.

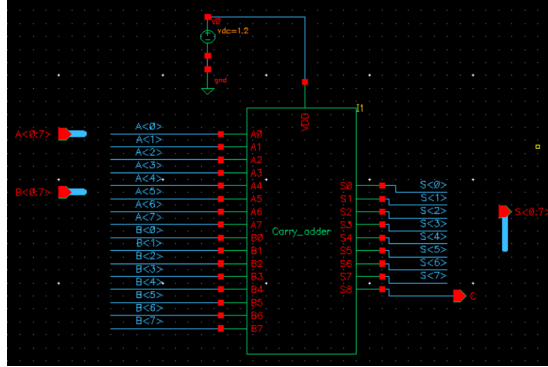


Figure 6: Final Compact Schematic for Easier Simulation and Power Calculation

4 Simulation

For testing purposes, hexadecimal notation is used to represent the inputs and expected outputs. The following conventions are applied:

- A : Input vector A
- B : Input vector B
- S : Expected sum output
- C : Expected final carry-out bit

Each test case is represented as a combination of $AABBSSCC$ in hexadecimal format for clarity and compactness.

BB	F5	B0	1
E9	3F	28	1
1F	49	68	0
F0	AF	9F	1
E5	C4	A9	1
CB	11	DC	0
EE	33	21	1
11	00	11	0
1A	A3	BD	0
C4	6B	2F	1
1F	F1	10	1
55	AC	01	1
C3	E2	A5	1
32	28	5A	0
FF	FF	FE	1

Figure 7: Input Vectors and Expected Output Values in Hexadecimal Format

To prove the functionality of the Manchester carry chain adder, the expected outputs are compared against the simulation results. The matching values confirm the correct operation of the design.

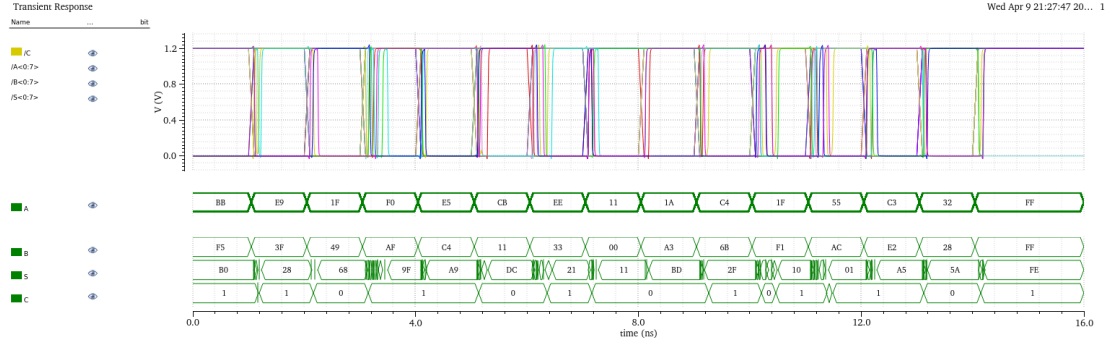


Figure 8: Simulation Results Showing Correct Functionality of the Manchester Carry Chain Adder

5 Power and Maximum Propagation Delay

To calculate the power consumption of the Manchester carry chain adder, we measure the average current drawn from the supply. The total power is then calculated by multiplying the average current with the supply voltage ($V_{DD} = 1.2V$) using the formula:

$$P = I_{avg} \times V_{DD}$$

The following figures show the current measurements used for the power calculation:

`average(1.2*(-IT("/V0/PLUS")))`

Figure 9: Calculator Method Used

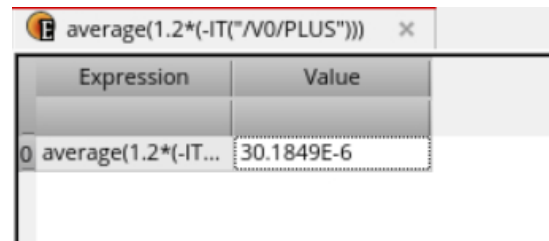


Figure 10: Final Result

From the simulation, the average power consumption of the Manchester carry chain adder is approximately $30 \mu W$.

We will now examine the maximum propagation delay of the Manchester carry chain adder. The maximum delay occurs when every carry bit is equal to 1, creating the worst-case scenario where the carry must propagate through the entire adder chain. In this situation, the longest critical path is from input bit A_0 to output bit S_7 .

To observe this, we initialize the input signals as follows:

- First input: $A = 80, B = 00$
- Then: $A = FF, B = 01$

These transitions are chosen to ensure that a carry must ripple through all stages. We then measure how long it takes for the outputs to switch and settle to their final steady state.

```
; data
80 00 00 0
ff 01 00 1
```

Figure 11: Input Vectors for Propagation Delay Measurement in Hexadecimal Format

The figure above shows the expected outputs of the additions between inputs A and B in the format $AABBSSCC$, similar to the previous input-output convention. In the next simulation, we will measure the maximum propagation delay based on these transitions.

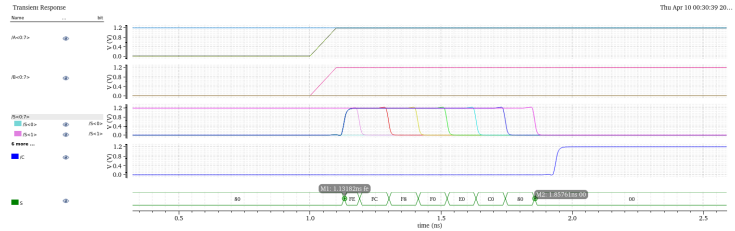


Figure 12: Simulation Results for Propagation Delay Measurement

The maximum propagation delay is calculated by taking the difference between the two measured times highlighted in the simulation. Specifically, the delay is:

$$PropagationDelay = 1.85761 ns - 1.13182 ns = 0.72579 ns$$

The throughput is simply the reciprocal of the propagation delay:

$$Throughput = \frac{1}{PropagationDelay} = \frac{1}{0.72579 \times 10^{-9}} \approx 1.378 \text{ GHz}$$

This means the Manchester carry chain adder can perform approximately 1.378 billion operations per second.

6 Comparison with Other Adders

There are various architectures suitable for high-speed addition. Here, we compare the Manchester carry chain adder with four parallel prefix-tree adders: Han-Carlson (HC), Brent-Kung (BK), Ladner-Fischer (LF), and Kogge-Stone (KS), across the criteria of layout complexity, logic depth, fan-out, and area.

Adder	Layout (Wiring Complexity)	Logic Depth	Fan-Out	Area
Manchester	Low (linear)	$O(n)$	Low	Low
Han-Carlson (HC)	Moderate	$O(\log n)$	Moderate	Moderate
Brent-Kung (BK)	Simple	$O(\log n)$	Low	Smallest
Ladner-Fischer (LF)	Complex	$O(\log n)$	High	Large
Kogge-Stone (KS)	Very Complex	$O(\log n)$	Low	Largest

Table 1: Comparison of Manchester Adder with HC, BK, LF, and KS Adders

As shown in the table:

- **Layout (Wiring Complexity):** The Manchester adder has simple, linear wiring. BK also has simple wiring. HC and LF have more complex layouts due to intermediate stages. KS has the most complicated and dense wiring network.
- **Logic Depth:** The Manchester adder has linear depth $O(n)$, making it slower for large bit-widths. All prefix adders (HC, BK, LF, KS) achieve $O(\log n)$ depth, making them faster for wide adders.
- **Fan-Out:** Manchester adder and BK adder have low fan-out. HC manages moderate fan-out. LF tends to have very high fan-out, while KS optimizes to maintain low fan-out even with high complexity.
- **Area:** Manchester adder consumes the least area due to simple chaining. BK is the most area-efficient among prefix adders. HC and LF require more area, while KS, despite being the fastest, occupies the largest silicon area.

Overall, the Manchester adder offers excellent area and layout simplicity, while prefix adders like KS and LF provide speed at the cost of higher complexity and area.